

Mathematics for Computing

Assignment 1

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August 13, 2026

Definition of a Set

What is a Set?

A set is a well-defined unordered collection of distinct elements.

Example: Positive odd integers less than 10

$$S = \{1, 3, 5, 7, 9\}$$

$$S = \{x \mid x \in \mathbb{Z}^+ \wedge (x < 10) \wedge (\exists k \in \mathbb{Z} : x = 2k + 1)\}$$

Some more examples

1. Perfect Squares

$$A = \{1, 4, 9, 16, 25, 36, 49\}$$

$$A = \{x \mid x \in \mathbb{Z}^+ \wedge (\exists n \in \mathbb{Z} : (1 \leq n \leq 7) \wedge (x = n^2))\}$$

2. Powers of 2

$$B = \{1, 2, 4, 8, 16, 32, 64\}$$

$$B = \{x \mid x \in \mathbb{Z}^+ \wedge (\exists n \in \mathbb{Z} : (0 \leq n \leq 6) \wedge (x = 2^n))\}$$

3. Prime Numbers in a Range

$$C = \{2, 3, 5, 7, 11, 13, 17, 19\}$$

$$C = \{x \mid x \in \mathbb{Z}^+ \wedge (1 < x < 20) \wedge$$

$$(\forall a, b \in \mathbb{Z}^+ : (x = ab) \implies (a = 1 \vee b = 1)))\}$$

4. Absolute Value Inequality

$$D = \{-1, 0, 1, 2, 3, 4\}$$

$$D = \{x \mid x \in \mathbb{Z} \wedge |2x - 3| \leq 5\}$$

5. Modulo Arithmetic

$$E = \{3, 8, 13, 18, 23\}$$

$$E = \{x \mid x \in \mathbb{Z}^+ \wedge (x < 25) \wedge (\exists k \in \mathbb{Z} : x = 5k + 3)\}$$

6. Divisors of a Number

$$F = \{1, 2, 3, 4, 6, 8, 12, 24\}$$

$$F = \{x \mid x \in \mathbb{Z}^+ \wedge (\exists k \in \mathbb{Z}^+ : 24 = xk)\}$$

7. Roots of a Polynomial

$$G = \{-2, 0, 2\}$$

$$G = \{x \mid x \in \mathbb{Z} \wedge x^3 - 4x = 0\}$$

8. Factorials

$$H = \{1, 2, 6, 24, 120\}$$

$$H = \{x \mid x \in \mathbb{Z}^+ \wedge (\exists n \in \mathbb{Z}^+ : (n \leq 5) \wedge (x = n!))\}$$

9. Boolean Cartesian Product

$$I = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$$
$$I = \{(x, y) \mid x \in \mathbb{Z} \wedge y \in \mathbb{Z} \wedge (0 \leq x \leq 1) \wedge (0 \leq y \leq 1)\}$$

10. Intersecting Conditions

$$J = \{10, 12, 14, 16, 18\}$$
$$J = \{x \mid x \in \mathbb{Z}^+ \wedge (10 \leq x < 20) \wedge (\exists k \in \mathbb{Z} : x = 2k)\}$$

Types of Sets (AI & DS Context)

1. Finite Set

A set with a strictly limited number of elements.

E.g: The discrete set of activation functions evaluated during hyperparameter tuning:

$$A_{func} = \{\text{ReLU}, \text{Sigmoid}, \text{Tanh}\}$$

2. Infinite Set

A set with an unlimited number of elements.

E.g: The continuous state space or weight space in a neural network:

$$W = \mathbb{R}^n$$

3. Countable Set

A set whose elements can be mapped 1-to-1 with natural numbers.

E.g: The sequential set of training epochs in a model:

$$E = \{1, 2, 3, \dots\}$$

Types of Sets (AI & DS Context)

4. Null (Empty) Set ($\emptyset$)

A set containing no elements.

E.g: The set of false positive predictions in a flawless, 100% precision classification model:

$$FP = \emptyset$$

5. Singleton Set

A set containing exactly one element.

E.g: The optimal weight vector θ^* that represents the single global minimum of a strictly convex loss function:

$$M = \{\theta^*\}$$

6. Cardinality ($|S|$)

The total number of unique elements in a set.

E.g: The dimensionality of a dataset. If a dataset has n unique features, the cardinality of the feature set F is:

$$|F| = n$$

Types of Sets (AI & DS Context)

7. Equal Sets ($A = B$)

Sets containing the exact same elements, regardless of order.

E.g: Defining unique class labels for a computer vision model in different orders yields equal sets.

$$C_1 = \{\text{Cat}, \text{Dog}\}$$

$$C_2 = \{\text{Dog}, \text{Cat}\}$$

$$\implies C_1 = C_2$$

8. Equivalent Sets ($|A| = |B|$)

Sets possessing the exact same cardinality (number of elements).

E.g: An autoencoder where the set of input layer neurons (I) maps one-to-one with the set of output layer neurons (O).

$$|I| = |O|$$

Types of Sets (AI & DS Context)

9. Subset ($A \subseteq B$)

A set where all its elements are contained within another set B .

E.g: In Support Vector Machines (SVM), the set of support vectors (SV) is a subset of the training data (D).

$$SV \subseteq D$$

10. Superset ($B \supseteq A$)

A set that contains all the elements of another set A .

E.g: A massive pre-training text corpus (C) acts as a superset containing all the text in a smaller fine-tuning dataset (F).

$$C \supseteq F$$

11. Universal Set (U)

The overarching set containing all objects under consideration.

E.g: In NLP, the Universal Set is the entire language vocabulary corpus from which words are drawn.

12. Power Set ($\mathcal{P}(S)$)

The set of all possible subsets of S , including the null set and S itself.

E.g: In exhaustive feature selection, if we have a small feature set $F = \{f_1, f_2\}$, the power set represents every possible combination of features a model could be trained on:

$$\mathcal{P}(F) = \{\emptyset, \{f_1\}, \{f_2\}, \{f_1, f_2\}\}$$

Note: If the dataset has n unique features ($|F| = n$), the cardinality of the power set is:

$$|\mathcal{P}(F)| = 2^n$$

This explains why exhaustive feature search is computationally expensive for large datasets!

Inclusion-Exclusion Principle

Definition

A counting technique used to determine the cardinality of the union of overlapping sets by adding the sizes of individual sets and subtracting the sizes of their intersections to correct for overcounting.

Equation (For 2 and 3 Sets)

For two sets A and B :

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For three sets A , B , and C :

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Inclusion-Exclusion Principle

General Form

For n finite sets $A_1, A_2, \dots, A_n$, the exact cardinality of their union is formulated as:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n-1} \left| \bigcap_{i=1}^n A_i \right|$$

E.g: In Machine Learning probability calculations, this principle is crucial when computing the joint likelihood of multiple overlapping features, ensuring that the intersection of intersecting user behaviors or text features isn't counted multiple times.

Set Operations

Definition

Set operations are used to construct new sets from existing ones. The most common operations are Union ($\cup$), Intersection ($\cap$), Difference ($-$), and Complement (A^c or A').

Example: Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$

$$\text{Union: } A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$\text{Intersection: } A \cap B = \{3, 4\}$$

$$\text{Difference: } A - B = \{1, 2\}$$

Definition

A binary relation R from a set A to a set B is a subset of the Cartesian product $A \times B$. If $(a, b) \in R$, we say a is related to b , often denoted as aRb .

Example: Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$.

The Cartesian product is:

$$A \times B = \{(1, x), (1, y), (2, x), (2, y), (3, x), (3, y)\}$$

A relation $R \subseteq A \times B$ could simply be defined as:

$$R = \{(1, x), (2, y), (3, x)\}$$

Equivalence Relations

Definition

A relation R on a set A is an **equivalence relation** if it satisfies three properties: it is reflexive, symmetric, and transitive. It partitions the set into disjoint subsets called equivalence classes.

Example: Congruence Modulo 3

Let R be a relation on integers $\mathbb{Z}$ where $aRb \iff a \equiv b \pmod{3}$.

- **Reflexive:** $a - a = 0$, which is divisible by 3.
- **Symmetric:** If $3 \mid (a - b)$, then $3 \mid (b - a)$.
- **Transitive:** If $3 \mid (a - b)$ and $3 \mid (b - c)$, then $3 \mid (a - c)$.

Equivalence classes formed: $[0] = \{\dots, -3, 0, 3, \dots\}$, $[1]$, and $[2]$.

Chains and Anti-chains

Chain

A subset of a partially ordered set (poset) is called a **chain** if every pair of elements in the subset is comparable (forming a total order).

Anti-chain

A subset of a poset is called an **anti-chain** if no two distinct elements in the subset are comparable to each other.

Example: Power set of $S = \{a, b, c\}$ under subset inclusion ($\subseteq$)

- **Chain:** $\{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}\}$
(Each element is a subset of the next)
- **Anti-chain:** $\{\{a\}, \{b\}, \{c\}\}$
(None of these are subsets of one another)

Partially Ordered Sets (Posets)

Definition

A partially ordered set (or poset) consists of a set S and a binary relation $\preceq$ that satisfies three properties: it is reflexive, antisymmetric, and transitive.

$$(S, \preceq)$$

Properties

For all elements $a, b, c \in S$:

$$a \preceq a \quad (\text{Reflexive})$$

$$(a \preceq b \wedge b \preceq a) \implies a = b \quad (\text{Antisymmetric})$$

$$(a \preceq b \wedge b \preceq c) \implies a \preceq c \quad (\text{Transitive})$$

Simple Example

The set of positive integers $\{1, 2, 3\}$ alongside the standard less-than-or-equal-to relation:

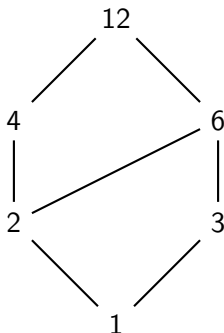
$$S = \{1, 2, 3\}, \quad \text{Relation: } \leq$$

Hasse Diagram 1: Divisibility

Example 1: Divisors of 12

Let S be the set of positive divisors of 12, and the relation $\preceq$ be “divides” (where $a \mid b$).

$$S = \{1, 2, 3, 4, 6, 12\}$$

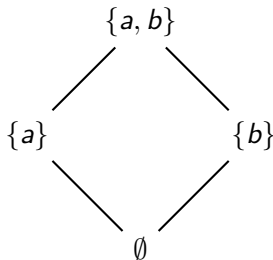


Hasse Diagram 2: Subset Relation

Example 2: Power Set of $\{a, b\}$

Let S be the power set of $\{a, b\}$, and the relation $\preceq$ be the subset inclusion ($\subseteq$).

$$S = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$



Hasse Diagram 3: Total Order (Chain)

Example 3: Less-than-or-equal-to

Let S be the first four natural numbers, and the relation $\preceq$ be the standard less-than-or-equal-to ($\leq$). This forms a strict chain because every element is comparable.

$$S = \{1, 2, 3, 4\}$$

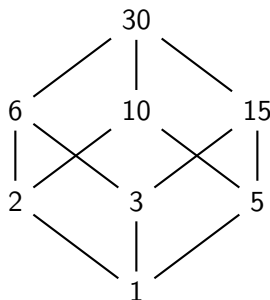


Hasse Diagram 4: Boolean Algebra (Cube Structure)

Example 4: Divisors of 30

Let S be the set of positive divisors of 30, and the relation $\preceq$ be “divides”. This creates a 3-dimensional cube structure in the Hasse Diagram.

$$S = \{1, 2, 3, 5, 6, 10, 15, 30\}$$



Hasse Diagram 5: Antichain

Example 5: Equality Relation

Let S be a set of discrete elements, and the relation $\preceq$ be strict equality ($=$). Because no two distinct elements are comparable, no edges are drawn.

$$S = \{a, b, c, d\}$$

a

b

c

d

Thank You!